

Q91. What does `{**{"a":1}, **{"b":2}, **{"a":10}}` return?

- a) `{"a":1,"b":2}` b) `{"a":10,"b":2}` c) `{"a":[1,10],"b":2}` d) Error - duplicate key

Q92. What is the Pythonic way to build a frequency counter in a plain dict?

- a) Check with if key in dict then increment b) Use `.get(key,0) + 1` c) Use `setdefault(key,0)` then increment d) Both B and C are valid

Q93. What does `frozenset({1,2,3}) | frozenset({3,4})` return?

- a) Error - frozensets are immutable b) `frozenset({1,2,3,4})` c) `[1,2,3,4]` d) `{1,2,3,4}`

Q94. What does `d={"a":1,"b":2}; list(d.items())` return?

- a) `["a","b"]` b) `[1,2]` c) `[("a",1),("b",2)]` d) `{"a":1,"b":2}`

Q95. What does `"a" in {"a":1, "b":2}` check?

- a) Whether "a" is in the values b) Whether "a" is in the keys c) Whether `("a",1)` is in items d) Error

Q96. What does `d.pop("key", None)` do when "key" doesn't exist?

- a) Raises `KeyError` b) Returns `None` c) Returns `{}` d) Does nothing (no return)

Q97. Which creates an empty dict, NOT an empty set?

- a) `set()` b) `{}` c) `frozenset()` d) `dict.fromkeys([])`

Q98. What does `max({"a":3, "b":1, "c":2})` return?

- a) 3 b) "c" c) "a" d) Error

Q99. What is the average time complexity of dict insertion?

- a) $O(n)$ b) $O(\log n)$ c) $O(1)$ amortized d) $O(n^2)$

Q100. What does `dict(zip("abc", [1,2,3]))` produce?

- a) `{"a":1,"b":2,"c":3}` b) `[("a",1),("b",2),("c",3)]` c) Error d) `{"abc":[1,2,3]}`

TOPIC 5: CONTROL FLOW & LOOPS (Q101–Q125)

Q101. What does the ternary expression `"pos" if x>0 else "neg"` evaluate to for `x=10`?

- a) "neg" b) "pos" c) True d) None

Q102. What does `list(range(5,0,-2))` produce?

- a) `[5,3,1]` b) `[5,4,3,2,1]` c) `[5,3,1,-1]` d) `[0,2,4]`

Q103. What prints when `continue` fires at `i==1` in `range(3)`?

- a) 0 1 2 b) 0 2 c) 1 2 d) 0

Q104. When does a for loop's else clause execute?

- a) When loop body raises exception
- b) When loop completes WITHOUT hitting break
- c) After every iteration
- d) When iterable is empty

Q105. In a while loop with break inside, does the else clause run?

`i=1; while i<5: i+=1; if i==3: break; else: print("done")`

- a) Yes, prints done
- b) No, break skips the else
- c) Yes, then prints 3
- d) Depends on the value of i

Q106. What does pass do in a loop body?

- a) Exits the loop
- b) Skips to next iteration
- c) Does nothing - a placeholder statement
- d) Pauses execution

Q107. What is total after: `total=0; for i in range(1,6): total+=i?`

- a) 10
- b) 15
- c) 5
- d) 6

Q108. What does `list(range(0))` return?

- a) [0]
- b) []
- c) Error
- d) None

Q109. What is the most Pythonic way to iterate over index AND value?

- a) `for i in range(len(data)): value=data[i]`
- b) `for i, value in enumerate(data):`
- c) `for i,value in zip(range(len(data)),data):`
- d) Both B and C are Pythonic

Q110. What does the nested loop produce?

`results=[]; [results.append(x*y) for x in [1,2,3] for y in [4,5]]`

- a) [4,5,8,10,12,15]
- b) [4,8,12,5,10,15]
- c) [(1,4),(1,5),...] tuples
- d) [4,5,8,10,12,15]

Q111. For `n=15`, what does "fizz" if `n%3==0` else "buzz" if `n%5==0` else `str(n)` return?

- a) "fizz"
- b) "buzz"
- c) "fizzbuzz"
- d) "15"

Q112. Which loop construct is cleanest for iterating a collection without needing index?

- a) while loop with counter
- b) `for i in range(len(c))`
- c) `for item in collection`
- d) All are equivalent

Q113. What prints for `x=5`?

```
if x>3: print("A")
elif x>4: print("B")
else: print("C")
```

- a) A
- b) B
- c) A and B
- d) C

Q114. What is the final value of i?

`i=1; while i<100: i*=2; if i>30: break`

- a) 32
- b) 64
- c) 16
- d) 30

Q115. What does `[i for i in range(10) if i%2==0 if i%3==0]` return?

a) [0,6] b) [0,2,3,6] c) [2,4,6,8] d) [3,6,9]

Q116. What does `list(map(str, [1,2,3]))` return?

a) [1,2,3] b) ["1","2","3"] c) "123" d) ["str(1)","str(2)","str(3)"]

Q117. What does `list(filter(None, [0,1,"","hello",None,3.14]))` return?

a) [0,"hello",3.14] b) [1,"hello",3.14] c) [0,1,"",None] d) [None]

Q118. What is `sum(i for i in range(10) if i%2==0)`?

a) 20 b) 25 c) 30 d) 0

Q119. What does the for-loop on empty list print?

`for i in []: print(i)`

`else: print("empty")`

a) Nothing b) "empty" c) Error d) 0

Q120. What does `match = "low" if x<5 else "mid" if x<10 else "high"` give for `x=3`?

a) "low" b) "mid" c) "high" d) Error

Q121. Which correctly iterates over a dict's key-value pairs?

a) `for key,value in d:` b) `for key in d.values()` c) `for key,value in d.items()` d) `for i in range(len(d))`

Q122. What does `[x**2 for x in range(10)][-1]` return?

a) 81 b) 100 c) 9 d) Error

Q123. Time complexity of searching a sorted vs unsorted list?

a) $O(\log n)$ vs $O(n)$ with binary search b) $O(n)$ vs $O(n)$ always for lists c) $O(1)$ vs $O(n)$ d) $O(\log n)$ vs $O(\log n)$

Q124. What does `any(x>5 for x in [1,3,7,2])` return?

a) True b) False c) 7 d) [7]

Q125. How many times does `(i==j)` trigger in two nested `range(5)` loops?

a) 25 b) 5 c) 10 d) 0

TOPIC 6: FUNCTIONS, LAMBDA & MODULES (Q126–Q150)

Q126. What does a function return if it has no return statement?

a) 0 b) "" c) None d) Error

Q127. What does `greet("Ahmad")` print if `def greet(name, msg="Hello"): return f'{msg}, {name}!'`?

a) ", Ahmad!" b) "Hello, Ahmad!" c) Error d) "greet, Ahmad!"

93	B	Frozensets support all set operations but return frozenset (for union/intersection) since they are immutable.
94	C	.items() returns dict_items view of (key,value) tuples. list() materialises it.
95	B	in on a dict checks keys only. Use ".values()" or ".items()" to check values/pairs.
96	B	pop() with a default argument returns the default instead of raising KeyError.
97	B	{ } creates an empty DICT. To create an empty set, use set() - not {}.disabled
98	C	Iterating over a dict yields keys. max("a","b","c") lexicographically = "c".
99	C	Hash tables have O(1) average insertion. Worst case (hash collisions) is O(n) but rare.
100	A	zip("abc",[1,2,3]) = [("a",1),("b",2),("c",3)]. dict() converts list of pairs to dict.
101	B	Python ternary: value_if_true if condition else value_if_false. x=10>0 is True so "pos".
102	A	Start=5, stop=0 (exclusive), step=-2: 5, 3, 1. Next would be -1 which is <= 0, so stop.
103	B	continue skips the rest of the loop body and moves to the next iteration. i=1 is skipped.
104	B	The else clause runs only if no break was encountered. It's an "exhausted normally" signal.
105	B	break also prevents the else clause

from executing in while loops.

106	C	pass is a no-op. It's used as a syntactic placeholder where a statement is required but nothing should happen.
107	B	$1+2+3+4+5 = 15$. Gauss formula: $n*(n+1)/2 = 5*6/2 = 15$.
108	B	range(0) produces an empty sequence. list() gives [].
109	B	enumerate() is the idiomatic Python way. Options A and C are clunky and less readable.
110	A	Outer loop x, inner loop y: $1*4, 1*5, 2*4, 2*5, 3*4, 3*5 = [4, 5, 8, 10, 12, 15]$.
111	A	Evaluates left to right: $n\%3==0$ is True for 15, so returns "fizz" immediately. Never checks %5.
112	C	Direct iteration is the most Pythonic, readable, and idiomatic. No len(), no index tracking.
113	A	First True condition executes and skips the rest. $x>3$ is True, so "A" prints. elif is never evaluated.
114	A	i doubles: 2,4,8,16,32. At 32, $32>30$ triggers break. Final $i=32$.
115	A	Both conditions must hold: divisible by 2 AND 3 = divisible by 6. In range(10): 0, 6.
116	B	map() applies the function to each element. $\text{str}(1)="1"$, $\text{str}(2)="2"$, $\text{str}(3)="3"$.
117	B	filter(None, ...) keeps truthy values: 0 is falsy, "" is falsy, None is falsy. Keeps 1,"hello",3.14.
118	A	Even numbers in range(10): 0,2,4,6,8. Sum = $0+2+4+6+8 = 20$.

119	B	When the iterable is empty, no break occurs, so the else clause still executes.
120	A	$x=3 < 5$ is True so returns "low" immediately. Nested ternary stops at the first True condition.
121	C	.items() yields (key,value) tuples. Iterating d directly yields only keys.
122	A	range(10) ends at 9. $9**2 = 81$. [-1] accesses the last element.
123	A	bisect.bisect_left() on sorted list = $O(\log n)$. Linear scan on unsorted = $O(n)$.
124	A	any() short-circuits: when it finds $7>5=True$, it returns True immediately.
125	B	$i=j$ at (0,0),(1,1),(2,2),(3,3),(4,4) - exactly 5 times. The diagonal of a 5×5 grid.
126	C	Python implicitly returns None when no return statement is reached.
127	B	msg has a default value of "Hello". When not provided, the default is used.
128	B	*args packs extra positional arguments into a TUPLE (not list). **kwargs packs keyword args into a dict.
129	A	1,2 go into args=(1,2): len=2. a=3,b=4 go into kwargs={"a":3,"b":4}: len=2.
130	B	print() is a side effect. return passes the value back so callers can use it. print returns None.
131	B	CRITICAL: mutable default arguments persist across calls. The same list is reused each call.
132	B	LEGB: Local, Enclosing, Global, Built-in. Python searches in this order for